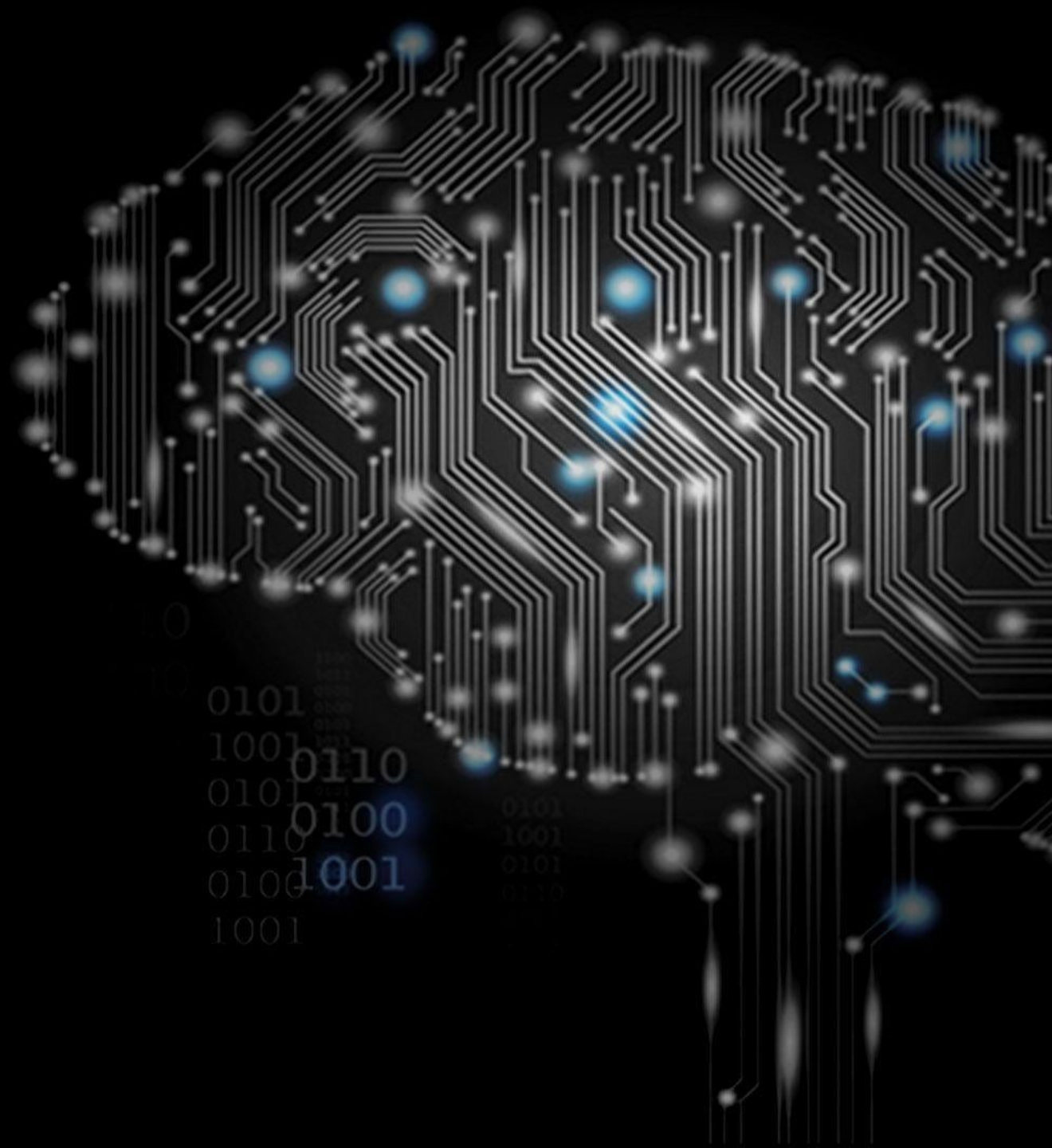


Introduction to Generative AI

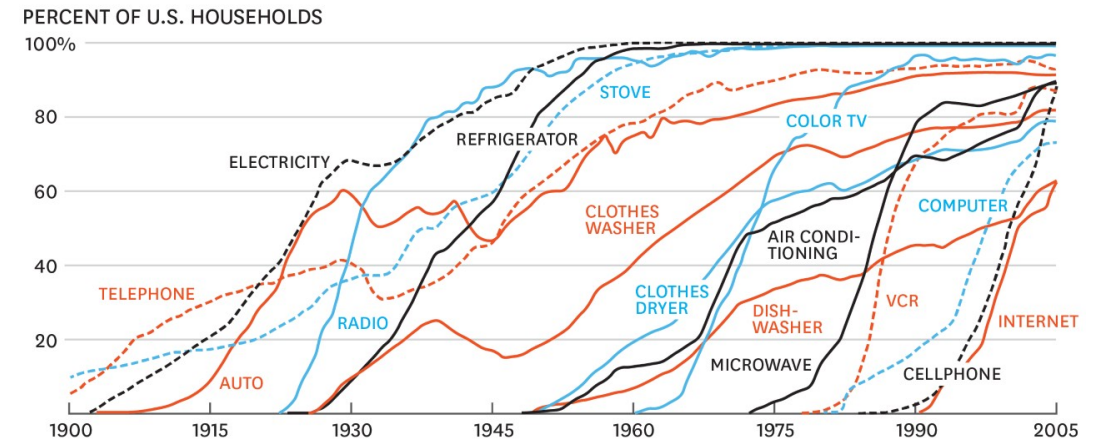
4. AI in the Real World



Disruptive Technology

- A disruptive technology is one that produces fundamental changes in the economy and society as a whole
- Tech is disruptive only when it reaches a “tipping point”
 - This happens when the technology is adopted by a critical mass of the society
 - This makes the development of a supporting infrastructure economically feasible
- Think of the evolution of the auto infrastructure, gas stations, roads, etc
 - A new technology may cause a burst of investment
 - But unless it reaches that tipping point, it will die out and be a classic bubble

CONSUMPTION SPREADS FASTER TODAY



SOURCE NICHOLAS FELTON, THE NEW YORK TIMES

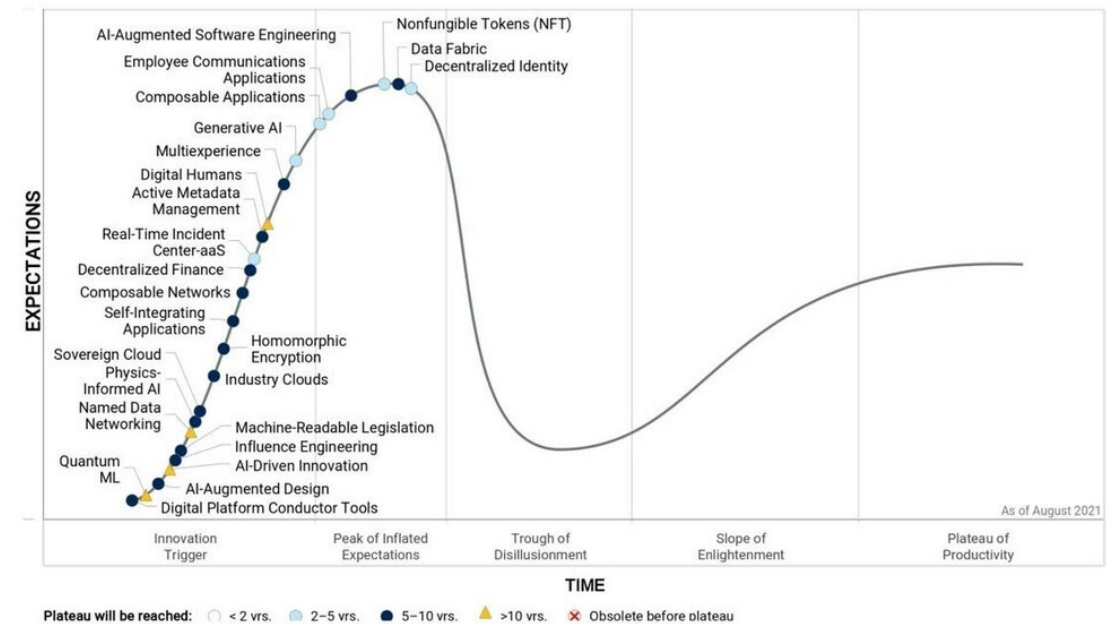
HBR.ORG



Technology Maturity

- Disruptive technologies and bubbles may both show an early explosion of interest and activity
 - This initial hype cycle induces massive investments
 - When the technology fails to deliver the promised hyped benefits, investments drop off
 - For a bubble, the investment die out
- For a disruptive technology, the initial hype bubble also declines
 - But there is enough return on investment that the technology continues to be adopted
 - Usually in more restricted forms, narrow areas of applications and limited scope
 - Hype is replaced with slower and more stable long-term growth
 - The dot com bubbled killed e-commerce in the 2000s but not the Internet
 - E-commerce slowly grew until it now dominates the retail marketplace

Hype Cycle for Emerging Technologies, 2021



Source: Gartner (August 2021)

747576



Case Study: Goldman Sachs' Financial Risk Management

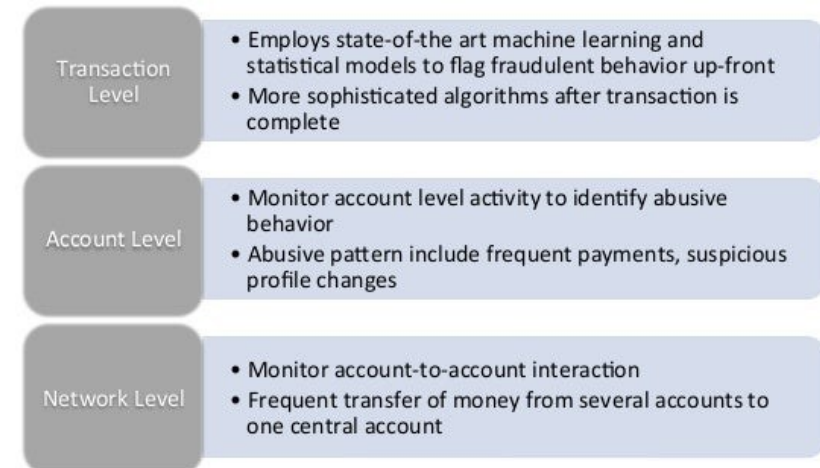
- Goldman Sachs needed an AI system to predict stock market movements and minimize investment risks.
- Solution:
 - Implemented machine learning models to analyze financial trends.
 - Used real-time AI monitoring to detect economic downturn warning signs.
 - Developed automated risk alerts for investment advisors.
- Outcomes
 - 25% reduction in financial risks associated with market volatility.
 - Faster, data-driven investment decisions.
 - Increased client portfolio profitability through AI-optimized strategies.



Case Study: PayPal's AI Fraud Detection System

- PayPal needed to prevent fraudulent transactions without disrupting legitimate payments.
- Solution
 - Developed an AI model that analyzed millions of transactions per second.
 - Used deep learning to detect patterns in fraudulent behavior.
 - Implemented real-time fraud prevention alerts for high-risk transactions.
- Outcomes:
 - Fraud detection accuracy improved by 85%.
 - Reduced false positives, ensuring smoother transactions.
 - Strengthened customer trust and financial security.

Fraud Prevention @ PayPal



Case Study: HSBC's Compliance and Fraud Detection System

- HSBC needed to ensure global compliance with financial data regulations and detect fraud.
- AI Strategy:
 - Deployed AI models to detect anomalies in financial transactions.
 - Used real-time compliance monitoring AI to track policy violations.
- Outcomes:
 - Reduced regulatory fines by 40% through proactive AI compliance.
 - Detected fraudulent transactions with 90% accuracy.



ROI of AI Initiatives

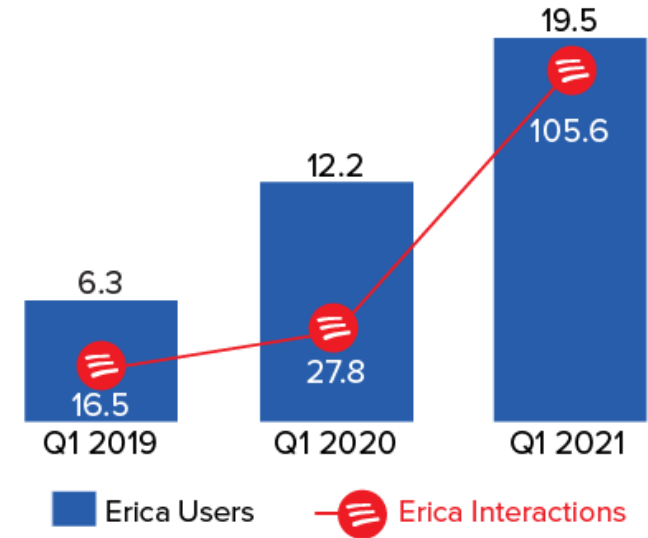
- How to Track AI Success:
 - Operational Efficiency: AI's impact on reducing costs and streamlining workflows.
 - Revenue Growth: AI's role in increasing sales, customer retention, or market expansion.
 - Workforce Productivity: Measuring how AI improves employee performance.
 - Customer Satisfaction: Analyzing AI's effect on customer engagement and service quality.
- AI Performance Metrics:
 - Time saved on manual processes.
 - Reduction in decision-making errors.
 - Increase in data-driven decision adoption rates.



AI Personal Assistants

- Using chatbots and other interactive tools allow for individualized assistance at a large scale and 24/7 responsiveness to queries and help requests
- Bank of America (Erica)
 - AI-powered chatbot "Erica" helps customers check account balances, track spending, and make payments
 - Uses natural language processing (NLP) to understand and respond conversationally

Total Erica Users and Interactions (MM)

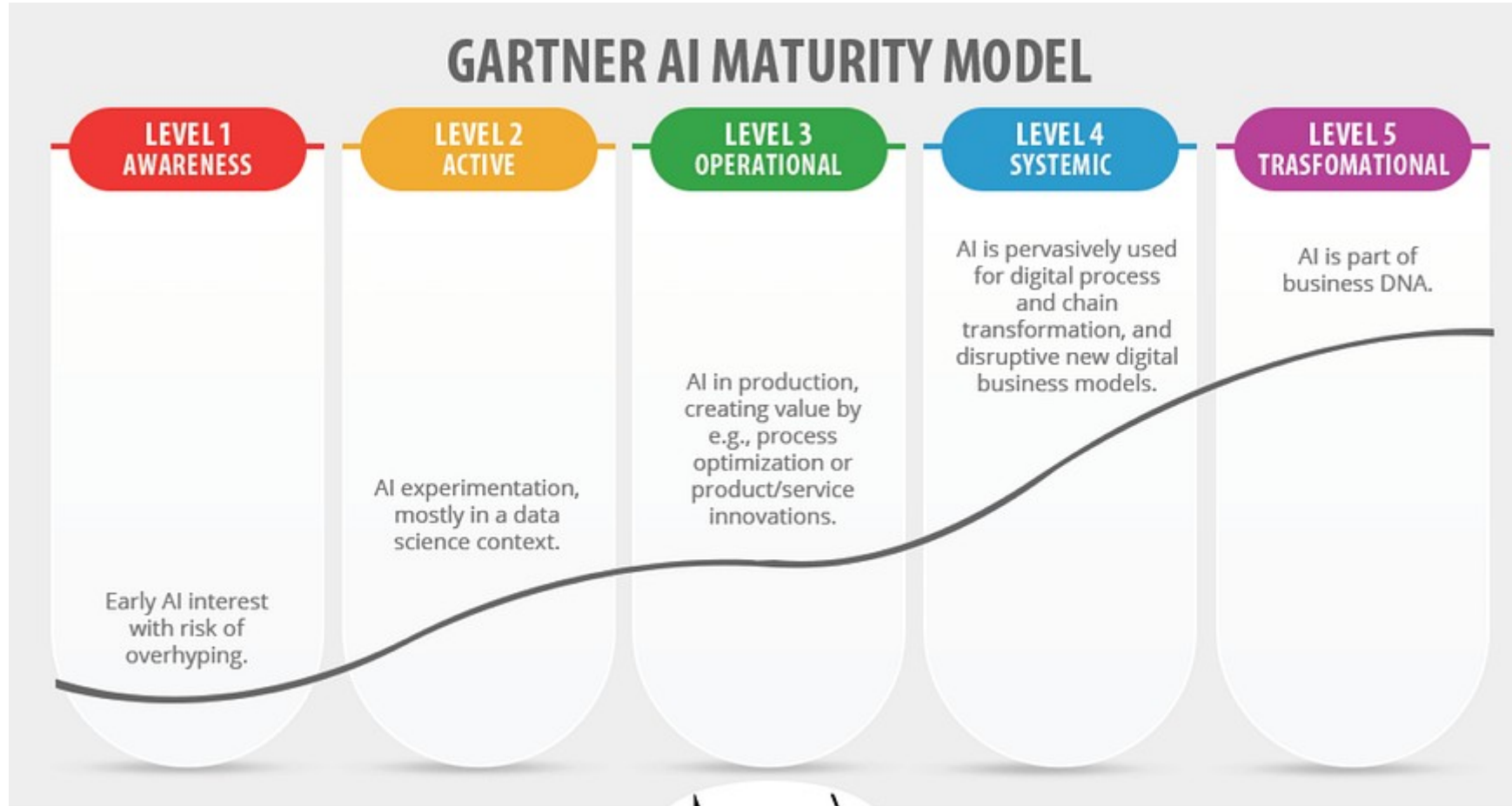


AI in Customer Support and Sentiment Analysis

- Improves customer service by analyzing customer feedback, predicting complaints, and automating responses
- Sentiment Analysis (Coca-Cola, Delta Airlines, etc.)
 - AI scans social media, emails, and customer reviews to gauge sentiment (positive, neutral, negative).
 - Helps brands proactively address negative feedback and enhance service quality.
- IBM Watson in Customer Support (Hilton Hotels – "Connie")
 - AI-powered concierge named "Connie" answers customer inquiries in hotels, providing recommendations and booking assistance.
 - Uses machine learning to improve responses over time using sentiment analysis.



Gartner's AI Maturity Model



Source: Gartner | Infographic design by Antonio Grasso

Datalogix

datalogix.blog



Tactical Innovation

- Common AI product targets
 - Improved use of data previously difficult to use
 - Q&A or summarization and searches of textual, image and other data
 - Automation of specific processes that are done manually
 - Introduction of new capabilities to extend or enhance existing operations
- Changes are often only visible in the organizations
 - External clients only see the results and improvements
 - They are unaware that AI is being used
 - Minimizes negative responses
 - Caveat: Using AI deceptively does produce harsh blow back when people find out



Facial Recognition... oops

MIT Technology Review

Artificial Intelligence Apr 8

...

New York's mass face recognition trial on drivers has been a spectacular failure

The trial: An internal e-mail from the Metropolitan Transportation Authority seen by the WSJ included the details of the trial at the Robert F. Kennedy Bridge last year. Cameras attached to the bridge were supposed to capture and identify the faces of drivers through their windshields as they passed, matching them against government databases.

The results: But the document, from November of last year, says that the "initial period for the proof of concept testing at the RFK for facial recognition has been completed and failed with no faces (0%) being detected within acceptable parameters." That's no faces accurately identified. Oops. Despite the failure, more cameras are going to be positioned on other bridges and tunnels, according to a spokesperson.



Facial Recognition... oops



Facial recognition technology has misidentified members of the public as potential criminals in 96 per cent of scans so far in London, new figures reveal.

The Metropolitan Police said the controversial software could help it hunt down wanted offenders and reduce violence, but critics have accused it of wasting public money and violating human rights.

Facial Recognition... oops



Facial recognition technology has misidentified members of the public as potential criminals in 96 per cent of scans so far in London, new figures reveal.

The Metropolitan Police said the controversial software could help it hunt down wanted offenders and reduce violence, but critics have accused it of wasting public money and violating human rights.

Apple News Summaries... oops

Business

Apple pulls AI-generated news summaries after feature repeatedly produced inaccurate headlines

Feature attributed wrong headlines to news organizations like BBC, Washington Post



[Jenna Benchetrit](#) · CBC News · Posted: Jan 17, 2025 10:54 AM EST | Last Updated: January 17



Apple News Summaries... oops

Business

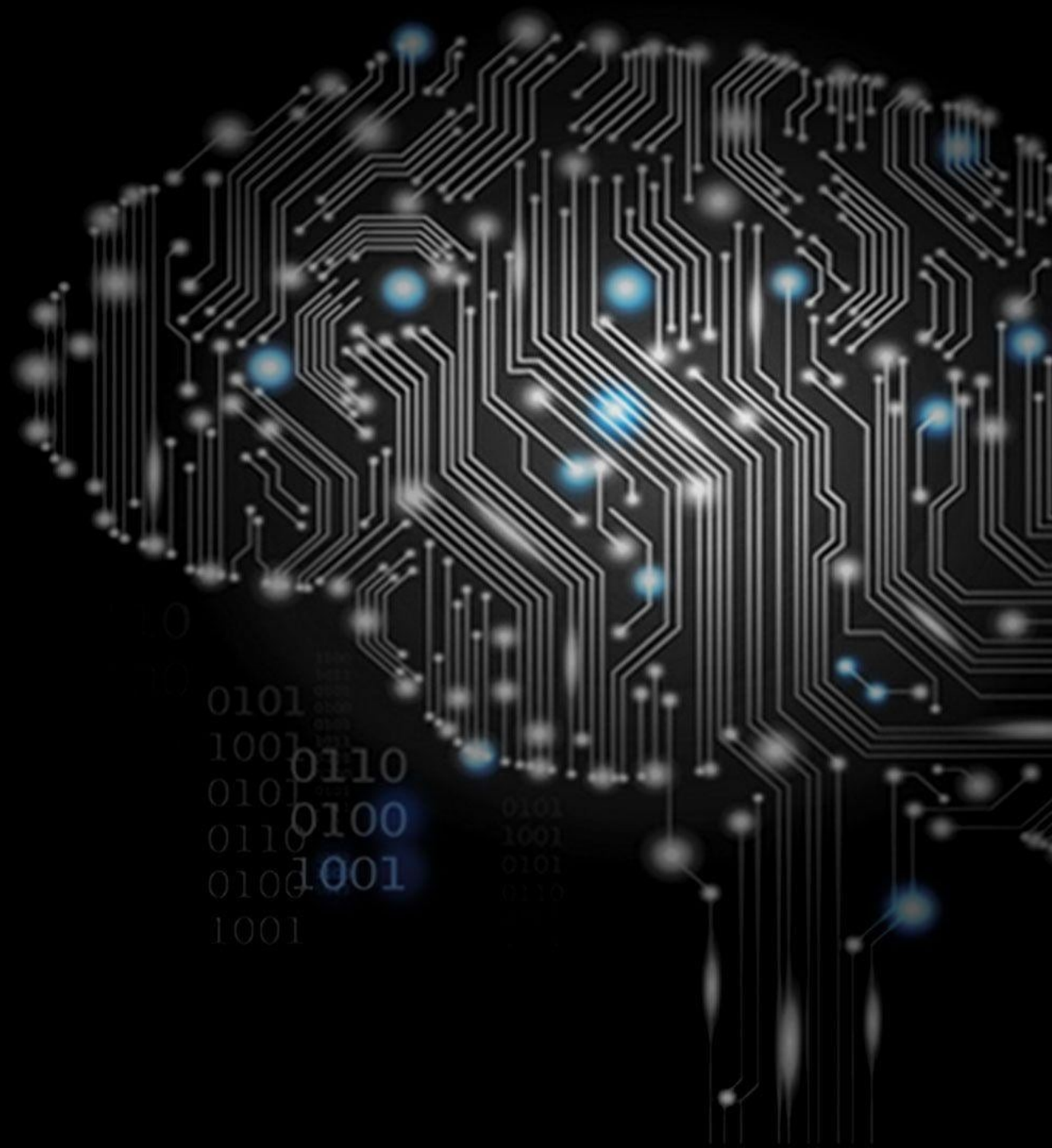
Apple pulls AI-generated news summaries after feature repeatedly produced inaccurate headlines

Feature attributed wrong headlines to news organizations like BBC, Washington Post



[Jenna Benchetrit](#) · CBC News · Posted: Jan 17, 2025 10:54 AM EST | Last Updated: January 17





0

0101
1001
0101
0110
0100
1001
1001

0101
1001
0101
0110